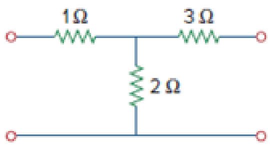
Chapter 19, Problem 3PP

Problem

Obtain the *y* parameters for the ***T*** network shown in Fig. 19.16.

Figure 19.16



Step-by-step solution

Step 1 of 10

Refer to Figure 19.16 in the text book.

Connect the current source $\mathbf{I}_1$ as shown in Figure 1. Identify the currents and voltages in the circuit.

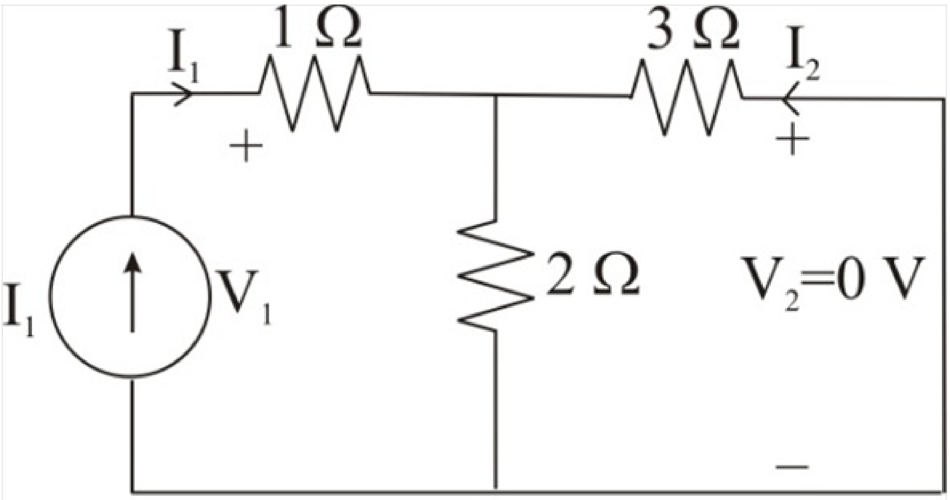


Figure 1

Step 2 of 10

Consider the expression for the voltage $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{V}_1 &= \mathbf{I}_1(1+2\|3)\\ &= \mathbf{I}_1\left(1+\frac{(2)(3)}{2+3}\right)\\ &= \mathbf{I}_1\left(1+\frac{6}{5}\right)\\ &= \mathbf{I}_1(1+1.2)\\ \mathbf{V}_1 &= \mathbf{I}_1(2.2)\end{aligned}$$

Therefore, the expression for the voltage $\mathbf{V}_1$ is $\mathbf{I}_1(2.2)$.

Step 3 of 10

Consider the expression for the current $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{I}_2 &= -\mathbf{I}_1\left(\frac{2}{2+3}\right)\\ &= -\mathbf{I}_1\left(\frac{2}{5}\right)\\ &= -\mathbf{I}_1(0.4)\end{aligned}$$

Therefore, the expression for the current $\mathbf{I}_2$ is $-\mathbf{I}_1(0.4)$.

Step 4 of 10

Determine the expression for the short circuit input admittance $\mathbf{y}_{11}$.

$$\mathbf{y}_{11} = \frac{\mathbf{I}_1}{\mathbf{V}_1}$$

Substitute $\mathbf{I}_1(2.2)$ for $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{Y}_{11} &= \frac{\mathbf{I}_1}{\mathbf{I}_1(2.2)}\\ &= 454.5 \text{ mS}\end{aligned}$$

Therefore, the expression for the short circuit input admittance $\mathbf{y}_{11}$ is $\boxed{454.5 \text{ mS}}$.

Step 5 of 10

Determine the value of the short circuit transfer admittance $\mathbf{y}_{21}$ from port-2 to port-1.

$$\mathbf{y}_{21} = \frac{\mathbf{I}_2}{\mathbf{V}_1}$$

Substitute $\mathbf{I}_1(2.2)$ for $\mathbf{V}_1$ and $-\mathbf{I}_1(0.4)$ for $\mathbf{I}_2$

$$\begin{aligned}\mathbf{y}_{21} &= \frac{-\mathbf{I}_1(0.4)}{\mathbf{I}_1(2.2)}\\ &= -181.82 \text{ mS}\end{aligned}$$

Therefore, the value of the short circuit transfer admittance $\mathbf{y}_{21}$ from port-2 to port-1 is $\boxed{-181.82 \text{ mS}}$.

Step 6 of 10

Connect the current source $\mathbf{I}_2$ as shown in Figure 2. Identify the currents and voltages in the circuit.

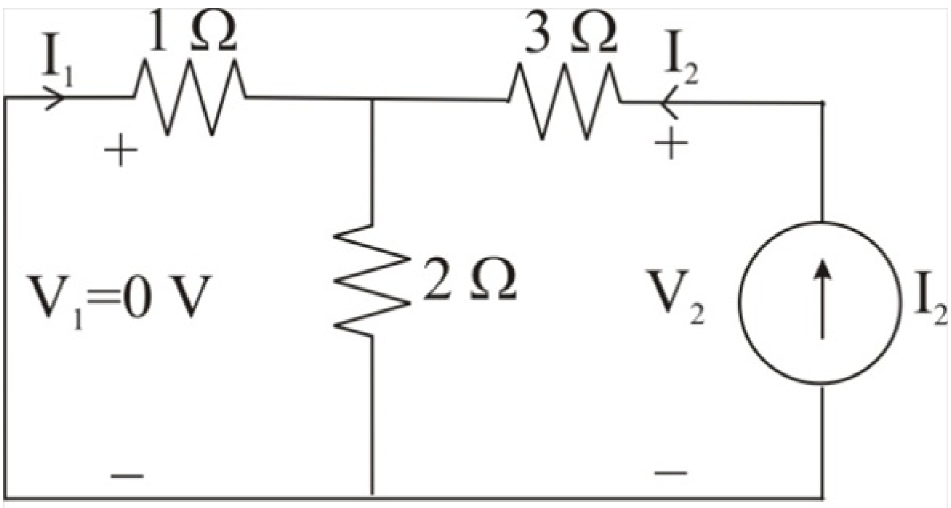


Figure 2

Step 7 of 10

Consider the expression for the voltage $\mathbf{V}_2$.

$$\mathbf{V}_2 = \mathbf{I}_2(3+2\|1)$$
$$= \mathbf{I}_2\left(3+\frac{(2)(1)}{2+1}\right)$$
$$= \mathbf{I}_2\left(3+\frac{2}{3}\right)$$
$$= \mathbf{I}_2(3+0.667)$$
$$\mathbf{V}_2 = \mathbf{I}_2(3.667)$$

Therefore, the expression for the voltage $\mathbf{V}_2$ is $\mathbf{I}_2(3.667)$.

Step 8 of 10

Consider the expression for the current $\mathbf{I}_1$.

$$\mathbf{I}_1 = -\mathbf{I}_2\left(\frac{2}{2+1}\right)$$
$$= -\mathbf{I}_1\left(\frac{2}{3}\right)$$
$$= -\mathbf{I}_1(0.667)$$

Therefore, the expression for the current $\mathbf{I}_1$ is $-\mathbf{I}_2(0.667)$.

Step 9 of 10

Determine the value of the short circuit transfer admittance $\mathbf{y}_{12}$ from port-1 to port-2.

$$\mathbf{y}_{12} = \frac{\mathbf{I}_1}{\mathbf{V}_2}$$

Substitute $\mathbf{I}_2(3.667)$ for $\mathbf{V}_2$ and $-\mathbf{I}_2(0.667)$ for $\mathbf{I}_1$.

$$\begin{aligned}\mathbf{y}_{12} &= \frac{\mathbf{I}_2(-0.667)}{\mathbf{I}_2(3.667)} \\ &= -181.81\ \text{mS}\end{aligned}$$

Therefore, the value of the short circuit transfer admittance $\mathbf{y}_{12}$ from port-1 to port-2 is $\boxed{-181.82\ \text{mS}}$.

Step 10 of 10

Determine the expression for the short circuit output admittance $\mathbf{y}_{22}$.

$$\mathbf{y}_{22} = \frac{\mathbf{I}_2}{\mathbf{V}_2}$$

Substitute $\mathbf{I}_2(3.667)$ for $\mathbf{V}_2$.

$$\begin{aligned}\mathbf{y}_{22} &= \frac{\mathbf{I}_2}{\mathbf{I}_2(3.667)} \\ &= 272.7\ \text{mS}\end{aligned}$$

Therefore, the expression for the short circuit output admittance $\mathbf{y}_{22}$ is $\boxed{272.7\ \text{mS}}$.